

EL OUED GUEMAR, Algeria

WMO: 605590

Lat: 33.511N

Lon: 6.777E

Elev: 62

StdP: 100.58

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	4.2	-5.2	2.5	14.4	-3.5	2.8	13.7	11.7	13.8	10.2	14.0	0.9	220	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	45.2	22.3	43.8	22.2	42.0	21.9	24.7	37.2	24.0	36.8	23.4	36.3	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.6	28.9	20.7	15.4	28.8	19.7	14.6	28.8	74.9	37.0	72.1	36.8	69.8	36.3	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.6	8.2	0.6	47.9	1.0	1.7	-0.1	49.1	-0.7	50.1	-1.3	51.0	-2.0	52.2		
(5)			-0.7	26.8	1.0	1.2	-1.4	27.7	-2.0	28.4	-2.6	29.0	-3.4	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	11.6	13.3	17.3	21.5	26.4	31.1	34.3	33.7	29.2	23.8	16.8	12.3
	DBStd	8.58	2.21	2.92	3.41	3.50	3.56	3.38	2.71	2.56	3.42	3.34	3.03	2.44
	HDD10.0	20	9	4	0	0	0	0	0	0	0	0	1	6
	HDD18.3	678	210	144	61	9	0	0	0	0	0	2	65	187
	CDD10.0	4638	58	97	226	345	508	634	754	733	575	426	204	79
	CDD18.3	2254	0	3	29	104	250	384	496	475	325	169	19	1
	CDH23.3	32984	1	35	327	1133	3171	5891	8383	7798	4400	1688	152	6
CDH26.7	20801	0	6	93	471	1744	3843	5959	5380	2554	724	27	0	
Wind	WSAvg	2.8	2.2	2.6	3.1	3.8	3.7	3.6	3.1	2.9	2.7	2.1	1.9	2.0
Precipitation	PrecAvg	72	19	4	7	7	3	1	1	3	9	6	6	8
	PrecMax	207	124	42	34	30	36	9	7	16	35	32	39	28
	PrecMin	25	1	0	0	0	0	0	0	0	2	0	0	1
	PrecStd	45	29	9	8	8	7	2	1	4	9	8	10	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	26.9	32.4	37.1	42.2	46.0	47.8	46.2	42.9	37.3	29.2	23.2
		MCWB	13.1	13.8	17.0	18.7	20.9	21.7	22.7	22.8	21.9	20.4	17.0	14.2
	2%	DB	20.1	24.0	29.1	34.0	39.1	43.9	45.9	44.8	40.3	35.0	26.9	21.0
		MCWB	12.2	13.6	15.3	17.4	20.3	21.5	22.7	22.4	21.9	20.0	16.6	13.0
	5%	DB	18.8	21.8	26.9	31.5	36.7	41.6	44.1	42.9	38.3	32.6	24.9	19.5
		MCWB	11.5	12.4	14.7	16.9	19.5	21.2	22.4	22.2	21.7	19.6	15.6	12.5
	10%	DB	17.2	19.8	24.8	29.2	34.6	39.2	42.1	41.0	36.2	30.6	23.1	18.1
		MCWB	10.9	11.6	14.0	16.2	19.1	20.9	22.2	22.2	21.7	19.3	14.8	11.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	16.0	18.6	20.4	23.2	24.7	26.0	25.7	25.3	23.2	19.2	15.7
		MCDB	20.1	25.0	27.7	31.4	36.2	38.2	39.4	37.7	34.6	31.5	27.0	20.4
	2%	WB	13.1	14.2	16.7	19.0	21.7	23.2	24.4	24.6	24.2	22.1	17.7	14.2
		MCDB	18.3	21.7	26.4	30.7	34.7	37.5	39.6	37.0	33.9	30.3	24.7	19.0
	5%	WB	12.1	13.1	15.5	17.8	20.6	22.3	23.6	24.0	23.4	21.1	16.4	13.3
		MCDB	17.4	20.2	25.0	28.7	33.8	36.9	39.2	36.6	33.0	28.8	23.2	18.1
	10%	WB	11.3	12.1	14.5	16.8	19.7	21.6	23.0	23.4	22.7	20.2	15.4	12.4
		MCDB	16.4	18.9	23.4	27.2	32.6	36.2	38.5	36.4	32.3	28.0	21.8	17.4
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.4	12.8	13.1	13.5	13.8	14.0	13.3	12.0	12.2	11.8	11.4
		MCDBR	13.3	14.3	15.5	16.0	15.9	16.4	16.4	15.5	14.2	14.3	13.2	12.6
		MCWBR	7.2	7.1	6.7	6.1	5.5	5.3	5.0	4.6	4.7	5.4	5.9	6.7
	5% WB	MCDBR	12.1	13.1	14.2	14.1	14.1	14.5	14.5	13.5	12.1	11.8	12.1	11.4
		MCWBR	7.2	7.1	6.8	6.2	6.0	5.7	5.3	4.7	4.7	5.0	6.2	6.7
Clear-Sky Solar Irradiance	taub		0.368	0.399	0.502	0.554	0.608	0.602	0.636	0.606	0.582	0.513	0.428	0.383
	taud		2.107	2.006	1.740	1.653	1.575	1.598	1.553	1.613	1.670	1.791	1.957	2.059
	Ebn at Noon		817	831	769	752	717	720	691	703	692	703	736	769
	Edh at Noon		125	154	218	249	272	266	277	256	232	190	145	125
All-Sky Solar Radiation	RadAvg		3.43	4.42	5.48	6.59	7.19	7.66	7.59	6.81	5.69	4.54	3.54	3.04
	RadStd		0.15	0.17	0.23	0.26	0.24	0.21	0.13	0.17	0.15	0.20	0.12	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.47	N/A	-0.90	N/A	N/A	N/A	-15	N/A	+132
(47) Regional (3 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	+113	+105

Nomenclature: See separate page